UNIT-1

Analytical Questions

① Cipher text:

PHHW PH DIWHU WKH WRJD SDUWB

key = 3

Plaintext:

MEET ME AFTER THE TOGA PARTY

② key = 11,

S→D, E→P, C→N, U→F, R→C, I→T, T→E, Y→J

ciphertext:

DPNFCTEJ

③

→ Only 25 possible keys

→ An Attacker tries every key

→ Time complexity : O(26) ≈ O(1) (constant time)

④

→ Letter frequencies remain unchanged

→ English letters like E, T, A occur most often

→ frequency analysis reveals the substitution

⑤ Ciphertext : GSRH RH Z HVXIVG

cipher : Atbash cipher

Plaintext : This is a secret

⑥ Most frequent ciphertext letter is X

→ Assume X → E first (since E is most common in English)

→ Then analyze common diagram (TH, HE, ER) and diagram (THE)

⑦ → one plaintext letter encrypts to different ciphertext letter

→ frequency distribution is flattened

→ frequency analysis become much harder

⑧ Key: KEYKEY KEYKEY

ciphertext: MVW2XM8VY2Lld

⑨ keyword: Lemon
plaintext: ATTACKATDAWN

⑩ → Repeated key creates repeating ciphertext patterns

→ kasiski examination estimate the key length from repeated sequences

→

(11) → It became a One-Time pad (OTD)

→ Each key is used only once and is truly random.

→ It provides perfect secrecy and cannot be broken by cryptanalysis if used correctly.

(12)

Monoalphabetic	Polyalphabetic
One substitution alphabet	Multiple substitution alphabet
Lower entropy	Higher entropy
Less confusion	Better confusion

(13) → Letter are represented as number (A=0, B=1, .. Z=25)

⇒ Encryption and decryption use mod 26 arithmetic

→ Example (Caesar): $C = (P+k) \bmod 26$

(14)

$$\begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$$

$H = 7, \ I = 8$

$C = K * P \pmod{26}$

$$\begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 8 \end{bmatrix} = \begin{bmatrix} 23 \\ 61 \end{bmatrix} = \begin{bmatrix} 23 \\ 9 \end{bmatrix} (\bmod 26)$$

$23 = X$

$9 = J$

(15) $\begin{bmatrix} 2 & 4 \\ 2 & 4 \end{bmatrix}$

- Determinant = $2 \times 4 - 2 \times 4 = 0$
- no inverse modulo 26
- decryption is impossible

(16)

→ The determinant must have a modular inverse

→ This is possible only if gcd (det, 26) = 1

→ otherwise, the key matrix is not invertible

(17)

→ Encrypt multiple letter together

→ Each ciphertext letter depends on several plaintext letter

(18)

→ A polyalphabetic substitution cipher (such as vigenere)

→ It hides normal english letter frequencies

(19)

→ plain: ABCDEFGHIJKLMNOPQRSTUVWXYZ

→ cipher: QWERTYUIOPASDFGHJKLZXCVBNM

(20)

→ using the same key, identical plaintext block produce identical cipher block

→ This can reveal repeating pattern making the cipher susceptible to pattern analysis

(21) $C = (7P + 3) \bmod 26$

- No
- It is an Affine cipher, not a Caesar cipher
- Caesar cipher: $C = (P + k) \bmod 26$
- $C = (aP + b) \bmod 26$, when $\gcd(a, 26) = 1$

(22)

- The attackers know some plaintext and its ciphertext
- By matching letter pairs they determine substitution

(23)

- Different key letter produce different substitution
- However, one plaintext letter affect only one ciphertext letter

(24)

- 'THE' is one of the most common English word
- Matching these pattern helps identify substitution for T, H and E

(25)

- Caesar cipher
- monoalphabetic cipher
- Vigner cipher
- Hill cipher

26) RETREATIMMEDIATELY does not match. The zigzag rail traversal was applied incorrectly.

27) Fill plaintext row-wise into a 4-column table and read column-wise. No padding is needed since ATTACKATDAWN has 12 letter

28) Repeating letter in the key create column numbering ambiguity. A fixed left-to-right rule is required to assign ranks.

29) No. the ciphertext does not match correct 2-rail zig-zag placement, indicating incorrect rail traversal.

30) Possible matrix : 4x6 (or 6x4). Since length = 24 no padding is required

31) key order for ZEBRA : 5, 3, 2, 4, 1. Rearrange columns in this order to recover the likely plaintext

32) frequency analysis fail because transposition changes position, not letter frequencies.

33) This is possible if different keys produce the Same column permutation or equivalent rearrangement.

34) Try decryption with rails 2,3,4 ... and choose the output that form meaningful plaintext.

35) Two keys of length 5 give $5! \times 5! =$ $120 \times 120 = 14,400$. possible key combination.

36) predictable padding (e.g. X) reveal message ending and helps attackers identify padding

37) Since 17 is prime, only 1X17 or 17X1 fit exactly. Other dimensions require padding.

38) Wrong filling assumption produces garbled plaintext, making decryption unsuccessful

39) In a 3 rail fence the first and last character remains on the outer rails so they often stay unchanged.

40) 23 characters with key length 6 → 24 cells needed, so 1 empty (padded cell

41) Keys: PLANT (5) and EARTH (5) ⇒ $5! \times 5! = 14{,}400$ possible permutations.

42) If all single-transposition attempts fail but double-transposition succeeds, it likely uses double transposition.

43) Repeating characters cause ranking ambiguity repeating - consistent left-to-right ordering

44) Periodic repetition every 5 letter is more likely columnar transposition than rail fence

45) Equal distribution across rows suggest the number of rail evenly divides the message length.

46) If the second key length divides the first, repeating pattern may weaken security.

47) Ciphertext become long only when padding is added to complete the final row or column.

48) A 6-column transposition has 6! = 720 possible keys. Brute force is practical

49) crypto CYTGAHRPORPY matches 2-rail Rail fence encryption of cryptography

50) Advantages: Easy ranking, no duplicate letter ambiguity

Disadvantages: Less memorable and may be easier to guess if poorly chosen